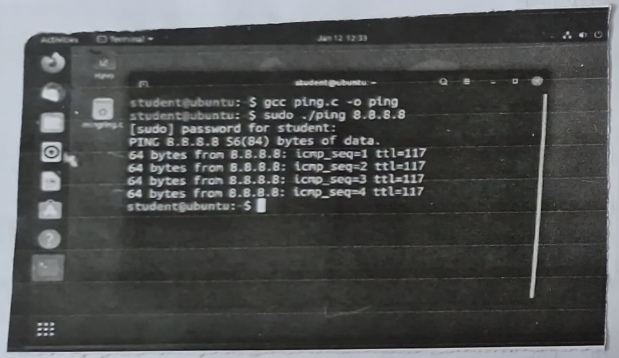


EXP-13 Implementing Ping program.

- Aim: To implement a custom ping program using RAW sockets in C to test network reachability by constructing and sending echo request and receiving ICMP reply packets.

- C-code implementation

```
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <netinet/ip_icmp.h>

unsigned short csum(unsigned short *buf, int nwords){
    unsigned long sum=0;
    for(; nwords>0; nwords--) sum+=*buf++;
    sum=(sum>>16)+(sum&0xffff);
    return ~sum;
}

int main(int argc, char *argv[]){
    if(argc!=2) return puts("Usage: sudo ./ping <ip>");

    int sock=socket(AF_INET, SOCK_RAW, IPPROTO_ICMP);
    struct sockaddr_in addr={.sin_family=AF_INET};
    inet_pton(...);

    struct icmphdr pkt={.type=ICMP_ECHO,
                        .code=0,
                        .un.echo.id=getpid()};

    for(int i=1; i<=4; i++){
        pkt.un.echo.sequence=i;
```

```
sendto(sock,&p,sizeof(p),0,(struct sockaddr*)&addr);
char buff[128];
socklen_t len = sizeof(addr);
int bytes = recv(sock, buf, sizeof(buf), 0);
struct iphdr *ip = (struct iphdr *)buf;
sleep(1);
}
close(sock);
return 0;
}
```

~~output:~~

~~gcc ping.c -o~~

- Result:

A custom 'ping' program was successfully implemented using RAW sockets and successfully executed and verified